Evidence-Bound Review Under a Fixed Deadline: A Reliability Contract for ICML-Style Review

Abstract

Time-bounded scientific review couples evidence quality with reliable batch execution. We present an anonymous Track 2 system that freezes paper identities, confines platform side effects to one Root coordinator, and is designed to use up to three paper-local Review Workers. Each Worker applies a private claim-map, falsification, score-anchoring, and consistency pass; deterministic validation precedes an optional, bounded verifier lane for compound high-risk drafts. On three 10-paper deterministic synthetic-mock runs, all 30 drafts reached mock verified completion with 100% schema validity and no duplicate post attempts or idempotency keys. A fault run recovered one malformed output, Worker timeout, claim timeout, and post timeout, and a separate two-process resume test preserved the four-paper ledger prefix and completed artifacts before reaching 10/10. A rule-based seeded-fixture check reached F1 1.0 versus 0.6667 for a one-finding baseline. These results validate local contracts and recovery logic only; they do not establish LLM review quality, human agreement, or live-platform throughput.

Serialized platform lane: discover, claim, reconcile, post, verify

↓ frozen manifest and bounded lease

Up to three Review Workers: paper-local evidence analysis only

↓ ReviewDraft

Deterministic validation → compound risk gate

Fast path or bounded read-only verifier → atomic ledger

Figure 1. Authority is narrow by construction. Workers cannot produce remote side effects or mutate shared state.

1. Problem and Contract

Track 2 requires an ICML-style review agent and a structured review of a frozen paper. The live operating window is 25 minutes, so review quality and batch reliability are coupled: a well-reasoned draft is not useful if it is lost, duplicated, or posted without confirmation. The system therefore treats evidence identity, review validity, and side-effect state as one end-to-end contract.

The official upstream skill is pinned at commit a9f4f258...153473f. Its Track 2-only path requires no new training, GPU, experiment service, or training metric. It requires frozen paper and evidence hashes, an independently frozen agent definition, a structured result, and explicit reporting when supplied evidence is insufficient. The local wrapper preserves the upstream Contribution assessment while retaining the event fields Significance, Originality, and Comment; it does not convert one field into another.

2. System Design

2.1. Single Owner for Side Effects

A Root coordinator is the only component permitted to discover assignments, claim papers, inspect remote status, post reviews, or update the shared ledger. The live design uses up to three persistent Review Workers. Each receives one frozen paper manifest at a time and returns only a structured ReviewDraft. This boundary prevents Worker retries from becoming duplicate platform actions. The synthetic-mock runtime models this bounded queue with deterministic local drafting; it is not a native-agent quality run.

2.2. Frozen Inputs and Evidence Trace

Before a worker starts, the coordinator records the paper identifier, paper file identity, SHA-256 and size, evidence file identities, hashes and sizes, aggregate input and evidence hashes, agent version, the canonical task-prompt asset hash, and the schema hash in a per-paper

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manifest. Native Worker instructions are frozen separately by the wrapper-package manifest, whose hash is recorded in batch metadata. The frozen corpus supplies allowed paths, and Root grants the active lease separately from the manifest. Unsafe paper identifiers are rejected before any artifact or ledger write. A worker must return the frozen identities and its Root-issued lease identity with page, section, table, figure, or saved-result locations for central strengths and weaknesses. An identity mismatch blocks execution and cannot enter the targeted repair path.

2.3. Canonical Review Shape

The draft contains Summary, Strengths, Weaknesses, Questions, Ethics and Limitations, Evidence Trace, and score rationales. It carries Soundness, Presentation, Contribution, Significance, Originality, Overall Recommendation, Confidence, and a constructive Comment as distinct values. A deterministic validator checks required fields, integer ranges, non-empty prose, lease and paper identity, evidence locations, and forbidden filler text. At most one targeted repair per paper is shared across schema and calibration defects; it may not change frozen identity.

2.4. Evidence-First Calibration

Before emitting JSON, a Worker privately builds a claim map, runs a falsification pass, anchors every score independently, and checks consistency across claims, rationales, scores, confidence, and comment. After deterministic validation, Root may route only a compound high-risk subset to one read-only verifier. The policy caps this lane at $\min(3, \lceil 0.3N \rceil)$ papers, 20 seconds per paper, before T+15 and only while validated backlog is at most two. Normal drafts use the fast path. These are execution-policy controls; they have not been evaluated as a review-quality improvement on unseen papers.

3. Operational Use

An operator first verifies the staged wrapper manifest, installs the two Skills and three native-agent definitions into project .codex, and starts a fresh Codex session. The invocation order is $auto-research, selecting its Track 2-only frozen-paper path, followed by $ralphthon-track2-review-agent. Root then freezes each assignment, dispatches paper-local drafting, runs deterministic validation and the bounded risk gate, and writes outbox and clipboard artifacts before any post attempt.

The bundled run_batch.py supports BUILD for manifest freezing and mock-only DRY-RUN for local recovery checks; it rejects LIVE before creating output. At the authorized live start, the Skill allows at most 45 seconds of read-only contract discovery. If assignment, field, or success-marker semantics cannot be verified, operation switches to the selector-free manual fallback while the same validation and status-first rules remain in force.

3.1. Operator Checklist

The executable handoff is deliberately short:

1. Run python3 scripts/install-track2-codex.py --check-staging, then rerun it with --install and open a fresh session so the staged Skills and native-agent definitions are loaded.

2. Invoke $auto-research and select its Track 2-only path; then invoke $ralphthon-track2-review-agent with exactly one of BUILD, DRY-RUN, or the authorized LIVE workflow.

3. For local preparation, call run_batch.py with --mode BUILD, --papers CORPUS, and --output-dir DIR. For a recovery rehearsal, change the mode to DRY-RUN and use --workers 3. Preserve the generated manifests, ledger, outbox, clipboard, and audit result.

4. During LIVE, do not use the bundled CLI. Root performs bounded read-only discovery, keeps all platform side effects serialized, and falls back to MANUAL_PLATFORM.md when the observed contract is incomplete. Stop only at verified remote completion or an explicit unresolved record.

4. Reliability Model

The coordinator uses a bounded queue with a default high-water mark of three, one active lease per paper, append-only ledger writes with atomic snapshot replacement, and idempotency keys for post intents. A timed-out worker may be re-queued once under the bounded retry policy while Root retains the deterministic lease. Malformed output records a validator-reject state and its validation errors before it can enter the shared targeted repair path.

Claim and post timeouts are ambiguous side effects. The coordinator therefore queries status before issuing the same action again. A post attempt remains post_unknown until remote state resolves it to verified success or a safe retry. The only live completion condition is posted_verified == assigned_count. Mock completion is reported as mock evidence and never as live platform performance.

5. Evaluation Protocol

Evaluation is split into quality and runtime contracts. Quality cases are seeded with issue type and evidence location before candidate generation. A fixed one-finding rule that selects the first explicit result sentence serves as the naive baseline. Deterministic matching uses paper ID, finding type, and normalized locator to report true positives, false positives, and false negatives. Finding text is retained for audit but is not part of the match, so this test does not evaluate semantic correctness. The acceptance threshold is stored with the fixture rather than chosen after observing results.

Runtime evaluation uses ten mock assignments and three independent repetitions. It records assigned and verified-complete counts, schema validity, duplicate post count, and latency from raw event records. Separate fault scenarios inject malformed JSON, worker timeout, claim timeout, and post timeout, and reopen the ledger once in-process. Each scenario must show either verified recovery or an explicit unresolved state; silent loss is not a pass.

6. Synthetic Mock Results

Table 1 reports only deterministic synthetic fixtures and the mock adapter. Across three normal runs, all 30 assigned paper instances reached mock posted_verified; all 30 drafts passed the schema, and duplicate post attempts and duplicate idempotency keys were both zero. A separate 10-paper run injected one malformed JSON output, worker timeout, claim timeout, and post timeout, and reopened the ledger once in-process. Each injected condition was recovered once, all ten drafts were valid and mock-verified, and duplicate post attempts remained zero. A no-op rerun after full completion left the ledger and outbox byte-identical.

OS-process resume behavior in the synthetic-mock runtime was tested separately. One process durably exited with code 75 after four mock-verified papers. A fresh process opened the same output directory and inputs, exited 0 at 10/10 mock verification with schema validity 10/10 and zero duplicate posts or keys, preserved the four-paper ledger prefix, and preserved the completed manifest and outbox bytes.

On the pre-frozen seeded synthetic issue set, the deterministic fixture-matching runtime emitted findings

Table 1. Deterministic synthetic-mock runtime evidence. No production claim or post occurred.

Run	Verified	Schema	Dup.
Normal, three repetitions	30/30	30/30	0
Fault injection	10/10	10/10	0
Two-process resume	10/10	10/10	0

Table 2. Seeded synthetic-issue quality evaluation.

System	TP	FP	FN	F1
Fixture-matching runtime	20	0	0	1.0000
One-finding rule	10	0	10	0.6667

at all 20 frozen type/location targets, yielding TP 20, FP 0, FN 0, and F1 1.0. The fixed one-finding baseline yielded TP 10, FP 0, FN 10, and F1 0.6667. Both had location accuracy 1.0 under the frozen rule. Since finding prose is not scored, these results test fixture type/location coverage rather than scientific judgment or semantic review quality.

Runtime and quality values come from the frozen evidence/mock-validation-final-20260712T1335KST/ aggregate.json. The actual process result comes from evidence/process-restart-proof/aggregate.json. Values above are direct aggregates or four-decimal rounding of their stored values.

A post-build policy evaluator additionally passed all 49 regression tests, confirmed exact installed/staged package equality, and repeated the normal synthetic-mock batch at 30/30 valid, duplicate-free completion. The native Worker instruction surface is 2,224 bytes under a 4,096-byte budget. This validates packaging and fast-path preservation, not the scientific effect of calibration.

7. Artifact and Audit Handoff

Each accepted local run leaves a self-contained evidence directory. Its manifests/ bind paper, evidence, prompt, schema, and agent identities; outbox/ and clipboard/ preserve validated review payloads before any post attempt; and ledger.jsonl, its atomic state snapshot, and summary.json record state transitions and completion counts. These artifacts are recovery inputs, not evidence of a remote submission.

A restart reuses the same output directory and frozen inputs. Already verified papers are skipped, but any manifest identity change is a hard failure rather than an overwrite. The receipt audit compares assigned paper IDs with posted_verified ledger states. In the live manual lane, Root records that state only after observing the platform success marker; an outbox file or local mock state is never promoted to a live receipt.

The final live audit begins at T+23:30, and new side effects stop at T+24:30. The handoff reports assigned, drafted, schema-valid, post_unknown, posted_verified, and unresolved counts together with an explicit mock or live label. A partial batch is therefore represented as an auditable unresolved result rather than silently counted as success.

8. Limitations

Passing mock runs cannot establish live platform throughput, submission success, review correctness on unseen research, or agreement with human judgment. The runtime elapsed times reflect in-process synthetic fixtures and are intentionally not used as a live speed claim. Worker-timeout evidence uses a mock adapter TimeoutError and bounded recovery; it does not prove that a running thread was forcibly killed.

Further limitations are structural. Evidence locations can show where a claim was derived but cannot guarantee the claim is correct. Originality assessment is limited to the submitted paper and its supplied references. A deterministic schema validator can reject malformed reviews but cannot certify scientific quality. The live adapter remains unspecified until read-only discovery reveals the actual platform contract; no endpoint or selector is assumed here. The evidence-first and verifier policies are prompt and orchestration contracts; their effect on unseen-paper quality and inter-reviewer agreement is unmeasured.

9. Conclusion

The design combines frozen evidence identity, a narrow side-effect owner, bounded paper-local Workers, private evidence-first calibration, deterministic validation, a compound risk gate, and status-first reconciliation. Its evaluation contract separates fixture coverage from semantic review quality and separates mock evidence from live outcomes. Deterministic mock runs satisfy the frozen runtime, type/location coverage, and two-process resume checks, while live platform behavior and unseen-paper review quality remain unmeasured.

References

Ralphthon ICML organizers. Auto Research Skill, Track 2-only frozen-paper path. Repository commit a9f4f2583648ef4ca54f980f951ae393d153473f, 2026.

Ralphthon ICML organizers. Track 2 rules, 2026.